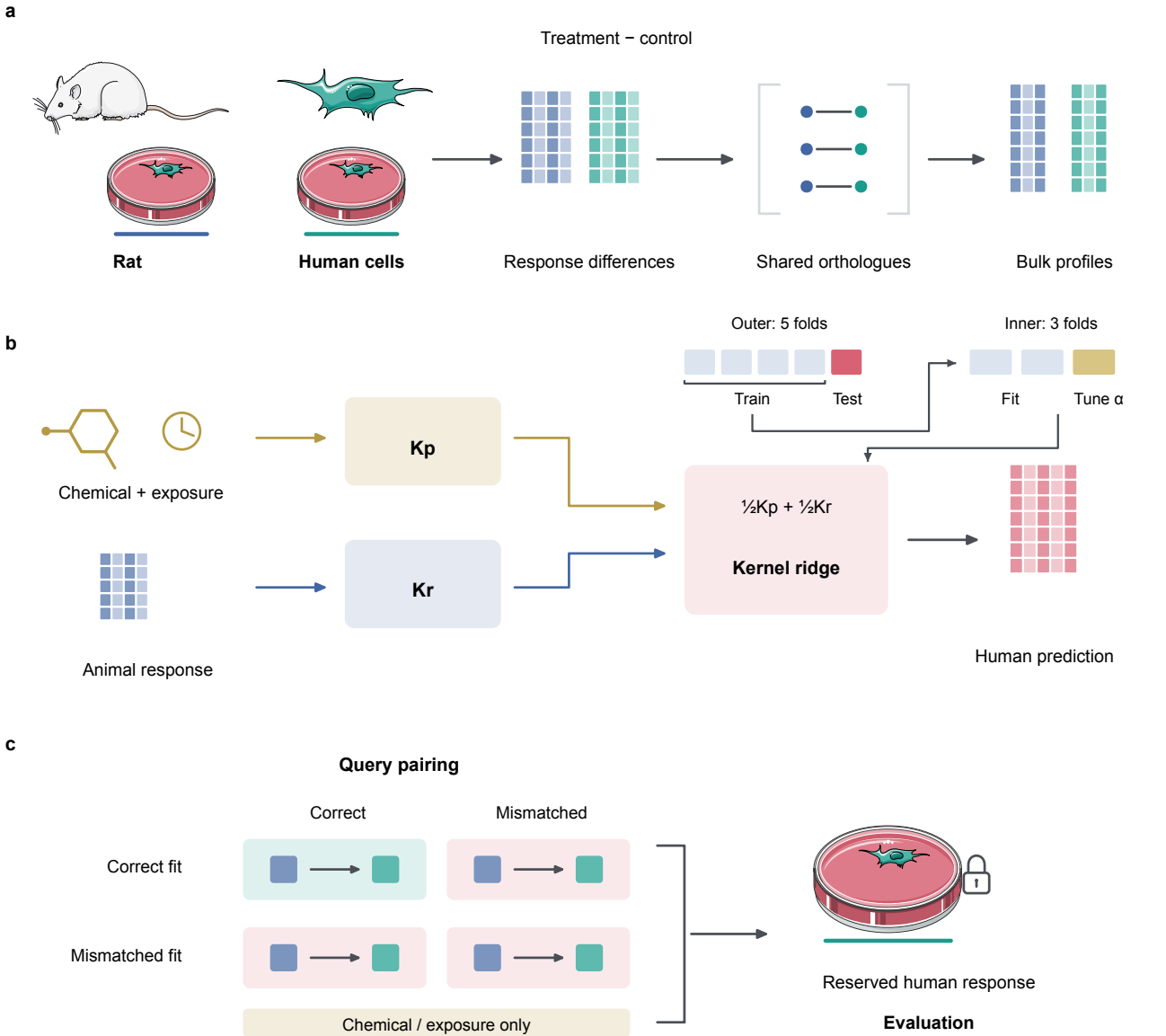
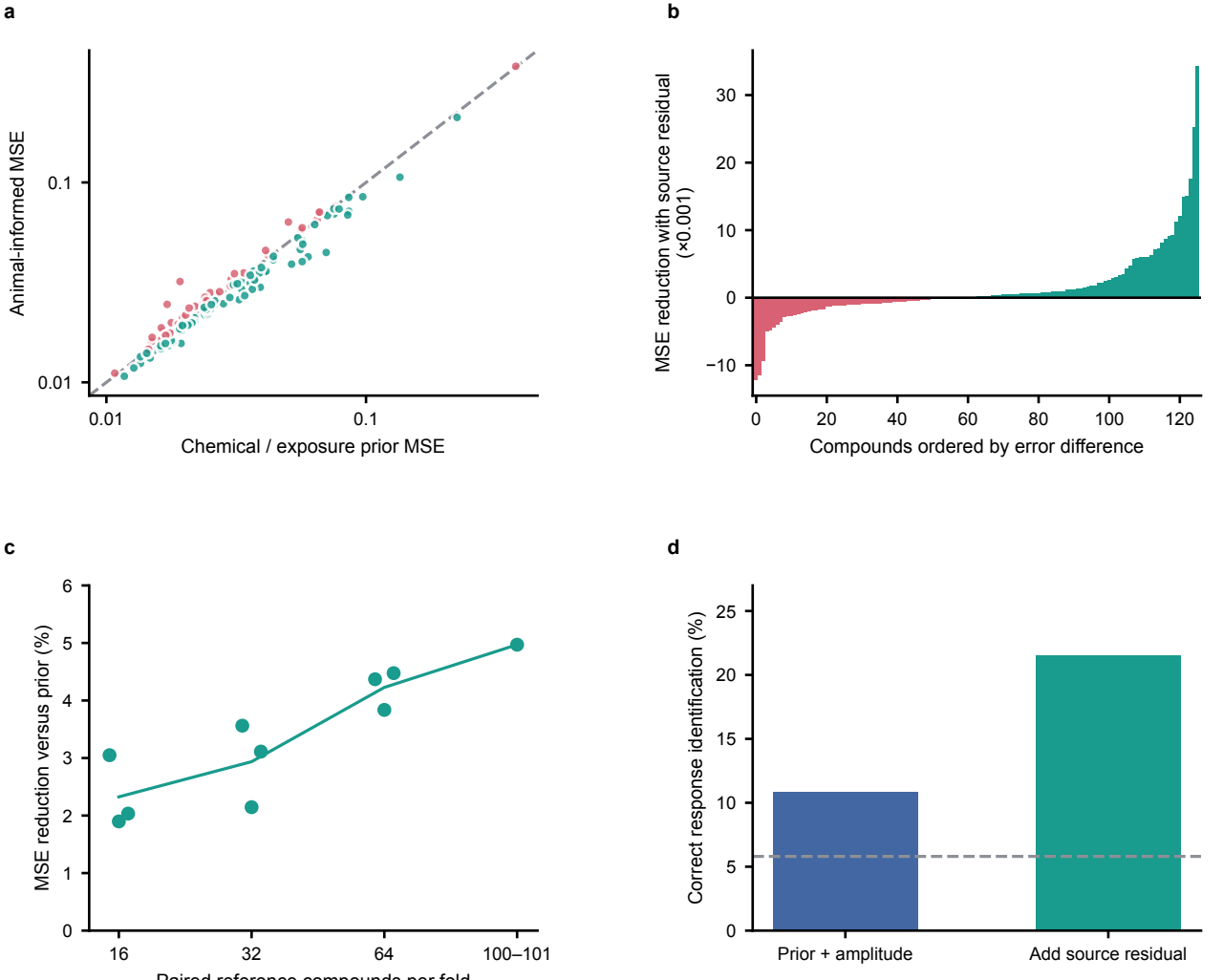
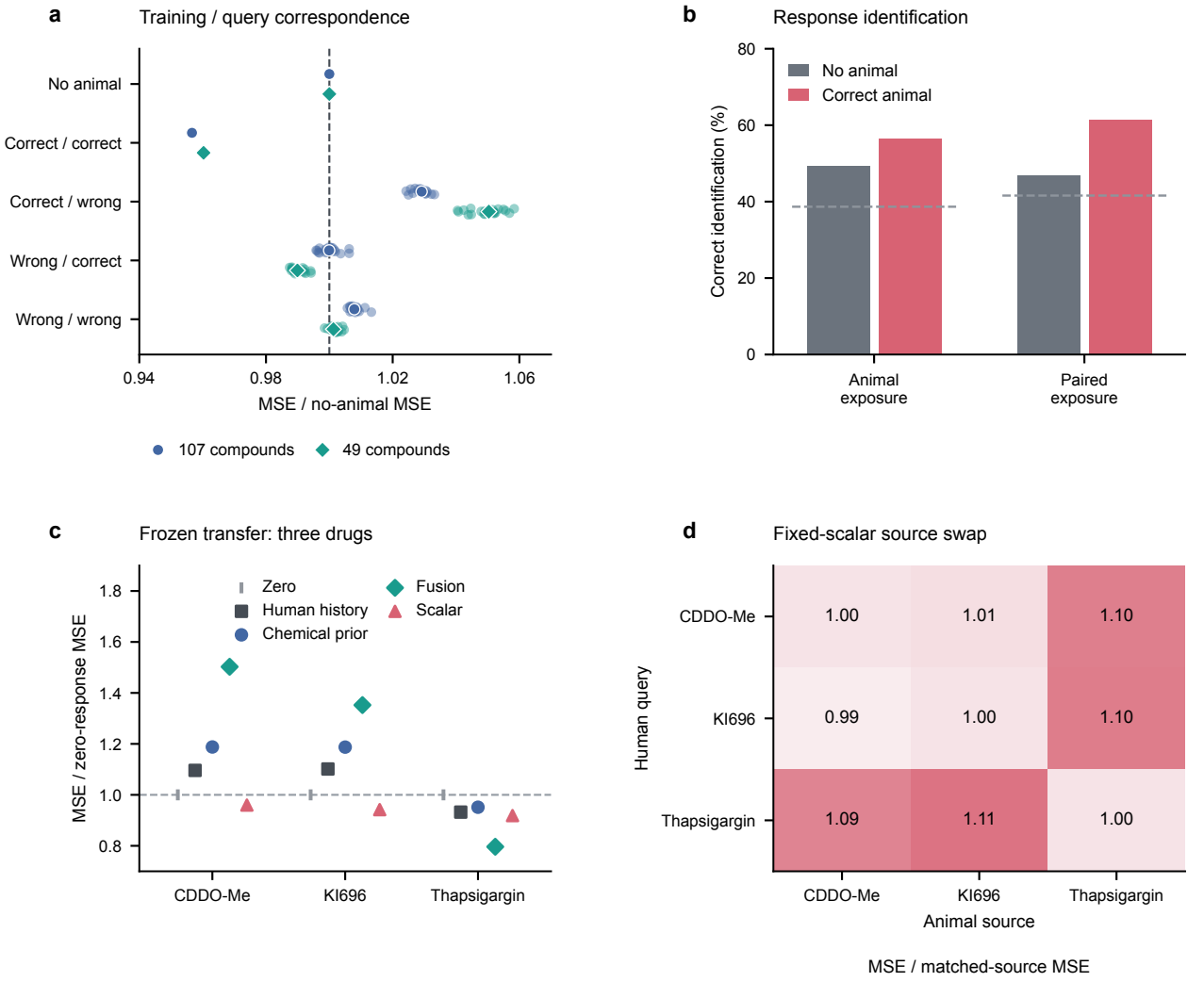
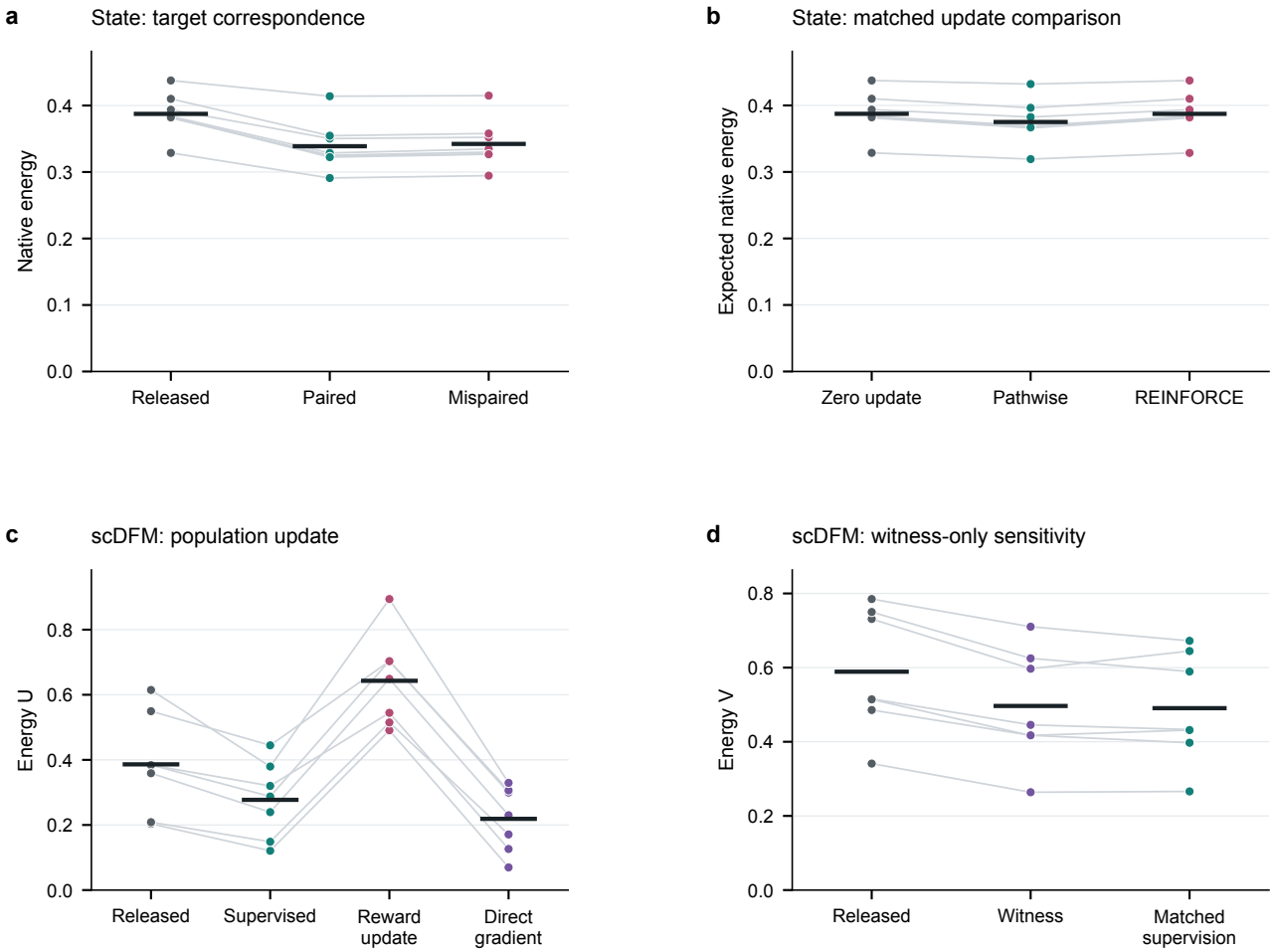
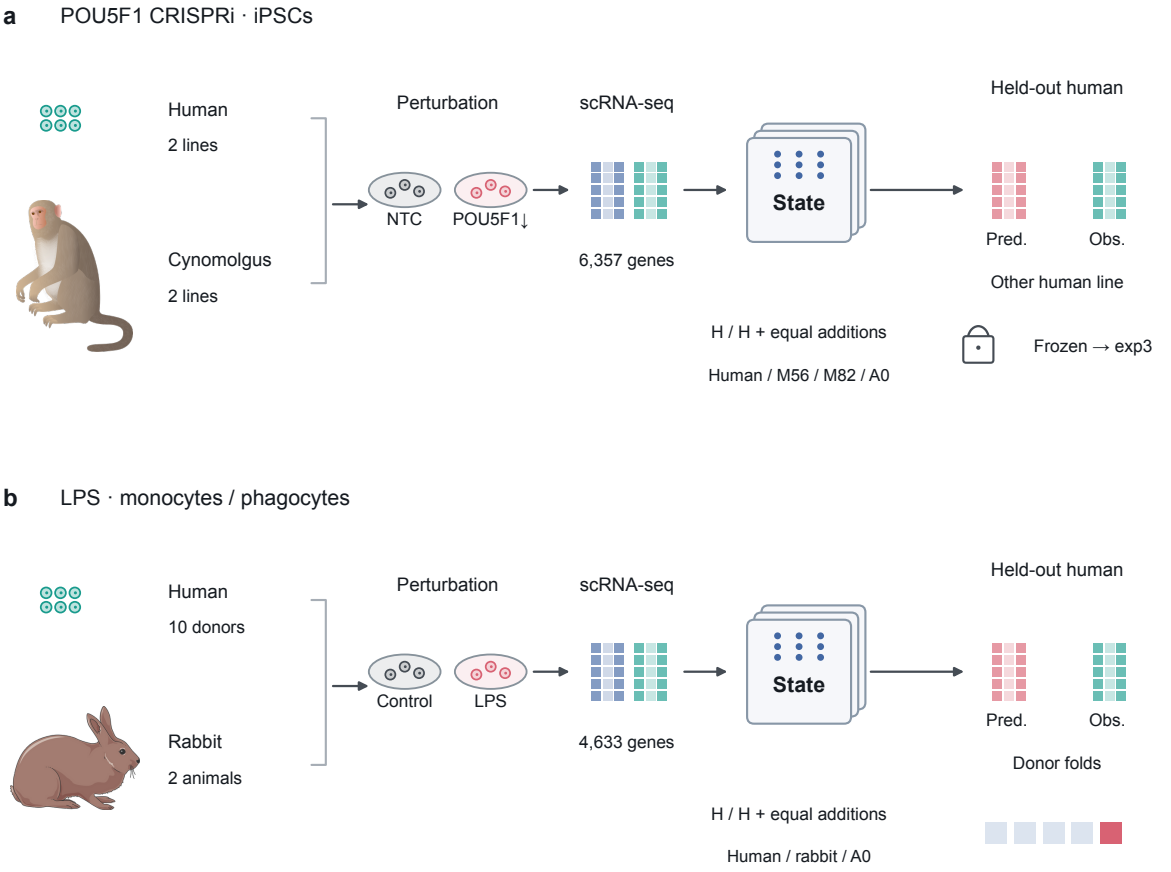
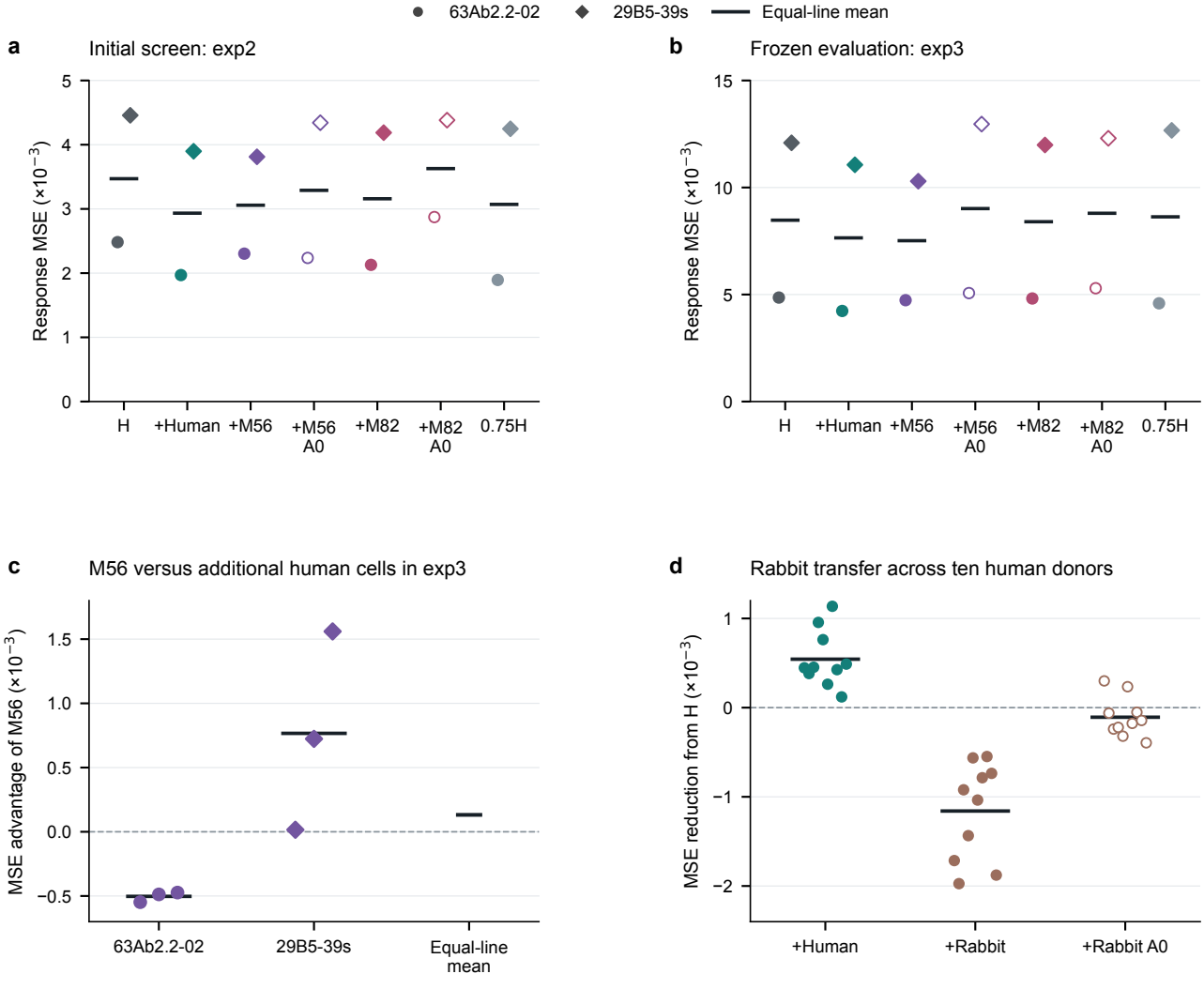

a
Treatment − control
Rat
Human cells
Response differences
Shared orthologues
Bulk profiles
b
Outer: 5 folds
Inner: 3 folds
Train
Test
Fit
Tune α
Chemical + exposure
Kp
Kr
½Kp + ½Kr
Kernel ridge
Animal response
Human prediction
c
Query pairing
Correct
Mismatched
Correct fit
Mismatched fit
Chemical / exposure only
Reserved human response
Evaluation


a
Animal-informed MSE
0.1
0.01
0.01
0.1
Chemical / exposure prior MSE
b
MSE reduction with source residual (×0.001)
30
20
10
0
−10
0
20
40
60
80
100
120
Compounds ordered by error difference
c
MSE reduction versus prior (%)
6
5
4
3
2
1
0
16
32
64
100–101
Paired reference compounds per fold
d
Correct response identification (%)
25
20
15
10
5
0
Prior + amplitude
Add source residual


a
Training / query correspondence
No animal
Correct / correct
Correct / wrong
Wrong / correct
Wrong / wrong
0.94
0.98
1.02
1.06
MSE / no-animal MSE
107 compounds
49 compounds
b
Response identification
No animal
Correct animal
Correct identification (%)
80
60
40
20
0
Animal exposure
Paired exposure
c
Frozen transfer: three drugs
Zero
Human history
Chemical prior
Fusion
Scalar
MSE / zero-response MSE
1.8
1.6
1.4
1.2
1.0
0.8
CDDO-Me
KI696
Thapsigargin
d
Fixed-scalar source swap
Human query
CDDO-Me
KI696
Thapsigargin
1.00
1.01
1.10
0.99
1.00
1.10
1.09
1.11
1.00
CDDO-Me
KI696
Thapsigargin
Animal source
MSE / matched-source MSE


a
State: target correspondence
Native energy
0.0
0.1
0.2
0.3
0.4
Released
Paired
Mispaired
b
State: matched update comparison
Expected native energy
0.0
0.1
0.2
0.3
0.4
Zero update
Pathwise
REINFORCE
c
scDFM: population update
Energy U
0.0
0.2
0.4
0.6
0.8
Released
Supervised
Reward update
Direct gradient
d
scDFM: witness-only sensitivity
Energy V
0.0
0.2
0.4
0.6
0.8
Released
Witness
Matched supervision


a
POU5F1 CRISPRi · iPSCs
Human
2 lines
Cynomolgus
2 lines
Perturbation
NTC
POU5F1↓
scRNA-seq
6,357 genes
State
Held-out human
Pred.
Obs.
Other human line
H / H + equal additions
Human / M56 / M82 / A0
Frozen → exp3
b
LPS · monocytes / phagocytes
Human
10 donors
Rabbit
2 animals
Perturbation
Control
LPS
scRNA-seq
4,633 genes
State
Held-out human
Pred.
Obs.
Donor folds
H / H + equal additions
Human / rabbit / A0


63Ab2.2-02
29B5-39s
Equal-line mean
a
Initial screen: exp2
Response MSE (×10⁻³)
H
+Human
+M56
+M56 A0
+M82
+M82 A0
0.75H
b
Frozen evaluation: exp3
c
M56 versus additional human cells in exp3
MSE advantage of M56 (×10⁻³)
63Ab2.2-02
29B5-39s
Equal-line mean
d
Rabbit transfer across ten human donors
MSE reduction from H (×10⁻³)
+Human
+Rabbit
+Rabbit A0